

Malicious URL Guard

Modern & Intelligent URL Security Analysis

ADVANCED CYBERSECURITY DECISION SUPPORT

A Decision Support Tool

Early-Warning Risk Assessment

Malicious URL Guard is designed as a **proactive decision support system** rather than a traditional antivirus.

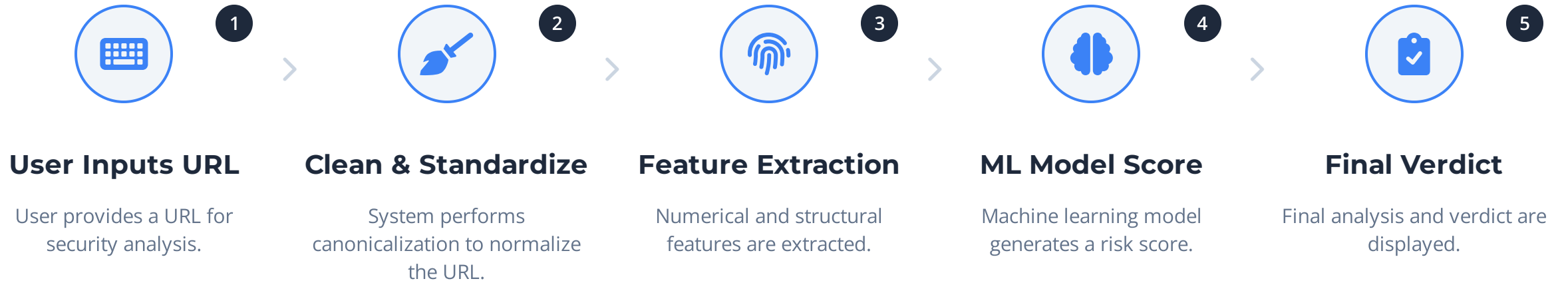
Structural Analysis

Evaluates threats based on structural properties of the URL (phishing, malware, defacement).

Security Reinforcement

Provides insights to strengthen security layers and prevent potential breaches.

System Workflow



Technical Components



Interactive Frontend

Built entirely in Python using [Streamlit](#) for a rapid, responsive UI without separate frameworks.



Data Engineering

[pandas](#) & [numpy](#) handle large-scale dataset manipulation, deduplication, and transformations.



Intelligence Engine

[scikit-learn](#) provides the algorithmic backbone (HistGradientBoostingClassifier), serialized via joblib.



Validation & Visualization

Automated testing via [pytest](#); dynamic visual metric reporting powered by [matplotlib](#).

Data Preparation & Vectorization



1. The Source

Kaggle Malicious URLs Dataset

Binary Classification: 0 (Benign) vs. 1 (Risky: Phishing, Malware, Defacement).

Trained on 588,792 rows of structural URL data.



2. Refinement

Canonicalization Process

Deduplicates records, removes invalid domains, and filters sensitive secrets.

```
http://Example.com:80/login/  
↓  
https://example.com/login
```



3. Vectorization

Feature Extraction

Converting structural patterns into numerical vectors for ML analysis.

String Lengths

Subdomain Counts

Symbol Ratios

IP Usage

Risky Extensions

The Hybrid Advantage

ML Engine

91.9%

ACCURACY

96.9%

ROC-AUC

Probabilistic ML identifying complex structural patterns across 588,792 data rows.

Rule Layer

Deterministic Failsafes

Hard-coded rules for non-negotiable security violations and blatant phishing behaviors like brand spoofing.

Synergistic Protection

Combining ML's ability to learn broad mathematical patterns with deterministic rules for a comprehensive security shield.